CCNA 200-301 | LAB 07

Corporate Network
FLSM Subnetting & Static Routing

Cisco Packet Tracer | Final implementation report | November 2, 2026

Scope: four equally sized IPv4 LAN subnets, two routers, one transit link, validated static and default routing, and evidence from the completed topology.

Executive summary

Designed and implemented four independent /26 departmental networks from 172.20.40.0/24. A two-router topology provides routed connectivity between all four LANs. The routers communicate over a separately allocated private /30 transit network. Cisco IOS routing tables and bidirectional ICMP tests demonstrate successful inter-subnet forwarding.

1. Requirements and subnet design

All four departments require a common fixed-length mask. /26 supports 62 usable host addresses per subnet, satisfying the largest requirement of 48 hosts. The four /26 blocks exhaust the assigned /24; the router-to-router transit link therefore uses a separate private block, 10.10.12.0/30.

Department	Required hosts	IPv4 subnet	Gateway	Usable hosts
Engineering	48	172.20.40.0/26	172.20.40.1	62
Administration	40	172.20.40.64/26	172.20.40.65	62
Commercial	36	172.20.40.128/26	172.20.40.129	62
Operations	30	172.20.40.192/26	172.20.40.193	62

2. Final topology and interface addressing

R1: G0/0 Engineering 172.20.40.1/26; G0/2 Commercial 172.20.40.129/26; G0/1 transit 10.10.12.1/30. R2: G0/0 Administration 172.20.40.65/26; G0/1 Operations 172.20.40.193/26; G0/2 transit 10.10.12.2/30. Each LAN connects two PCs through one access switch. The switches remain in independent physical LANs; VLAN configuration is outside the scope of this lab.

LAN	Host A	Host B	Gateway
Engineering	172.20.40.10	172.20.40.20	172.20.40.1
Administration	172.20.40.70	172.20.40.80	172.20.40.65
Commercial	172.20.40.140	172.20.40.150	172.20.40.129
Operations	172.20.40.200	172.20.40.210	172.20.40.193

3. Routing strategy

R1 has a specific static route to Administration (172.20.40.64/26) via R2 (10.10.12.2) and a default route via the same next hop. R2 has a specific static route to Engineering (172.20.40.0/26) via R1 (10.10.12.1) and a default route via R1. Routes to directly connected LANs are installed automatically when interfaces are operational. The default routes cover remaining remote LANs, while longest-prefix match selects any more-specific route.

R1

```
ip route 172.20.40.64 255.255.255.192 10.10.12.2
ip route 0.0.0.0 0.0.0.0 10.10.12.2
```

R2

```
ip route 172.20.40.0 255.255.255.192 10.10.12.1
ip route 0.0.0.0 0.0.0.0 10.10.12.1
```

The symmetrical default-route design is valid for the demonstrated LAN destinations. In a larger deployment, reciprocal defaults require care to avoid forwarding loops for unknown networks.

4. Validation results

The captured Cisco IOS outputs confirm connected (C), local (L), static (S), and candidate default (S*) entries on both routers. The displayed PC tests show successful ICMP Echo Replies between remote departments, including Operations to Engineering and Commercial, and Engineering to Administration and Commercial. Some initial pings show one or two timeouts, followed by successful replies; ARP resolution is a plausible cause, but the screenshots alone do not establish the exact cause.

Evidence	Observed result	Interpretation
PC0 to 172.20.40.80	2 replies / 4	Reachability confirmed
PC0 to 172.20.40.150	3 replies / 4	Reachability confirmed
PC7 to 172.20.40.10	3 replies / 4	Return routing confirmed
PC7 to 172.20.40.150	3 replies / 4	Cross-router forwarding confirmed

5. Engineering decision / troubleshooting

Hardware capability assessment: the available integrated routed ports on the original router did not support four separate Ethernet LAN connections plus transit. The completed implementation uses two routers with three routed GigabitEthernet interfaces each, avoiding reliance on switched expansion-module ports for routed interface addressing.

Address plan constraint: four equal /26 allocations consume the full 172.20.40.0/24 block. The separate private 10.10.12.0/30 point-to-point network preserves the departmental FLSM plan without overlap. Verification uses show ip interface brief, show ip route, and ICMP pings to confirm the final implementation.

6. Reproduction and evidence

Open the companion .pkt file when available, verify IPv4 masks and default gateways on the eight hosts, inspect R1/R2 via show ip route, and repeat cross-subnet ping tests. The following pages reproduce the eight screenshots supplied from the final lab session.

7. Screenshot evidence

Figure 1. Final corporate topology: four /26 departments connected through two routers and a /30 transit link.

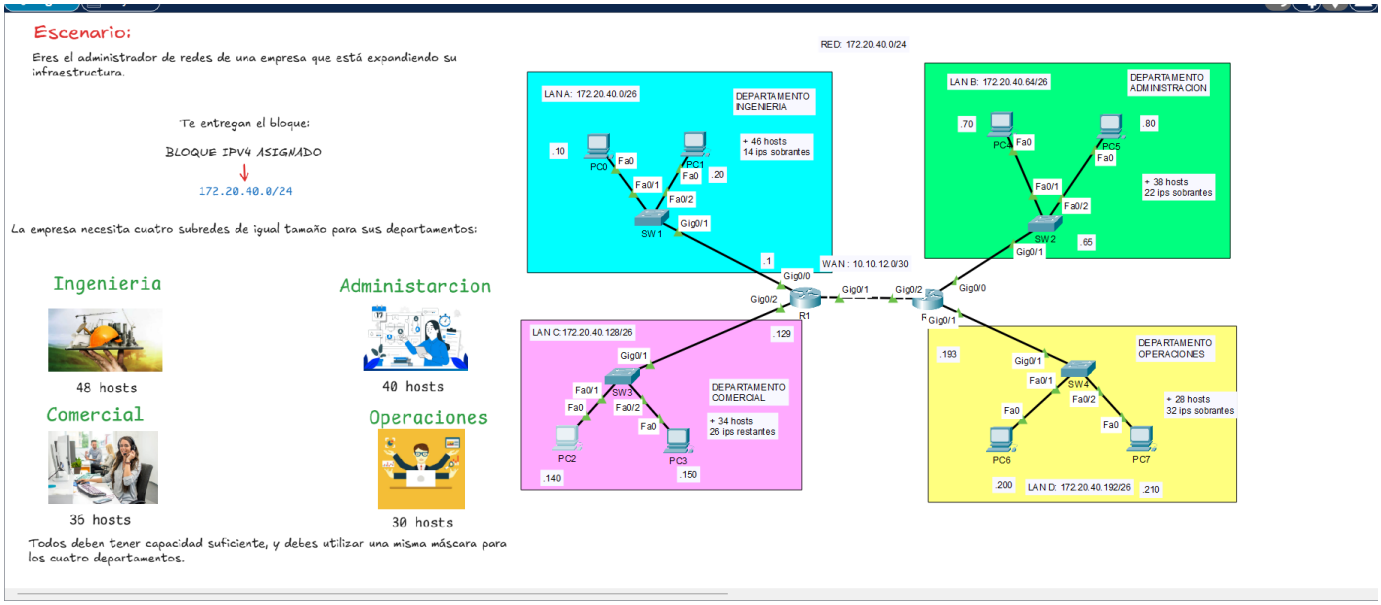


Figure 1. Final corporate topology: four /26 departments connected through two routers and a /30 transit link.

Figure 2. ICMP responses to Administration host 172.20.40.80 from the Engineering side.

```
C:\>ping 172.20.40.80

Pinging 172.20.40.80 with 32 bytes of data:

Request timed out.
Request timed out.
Reply from 172.20.40.80: bytes=32 time<1ms TTL=126
Reply from 172.20.40.80: bytes=32 time<1ms TTL=126

Ping statistics for 172.20.40.80:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Figure 2. ICMP responses to Administration host 172.20.40.80 from the Engineering side.

Figure 3. R1 saved configuration inspection (initial CLI output).

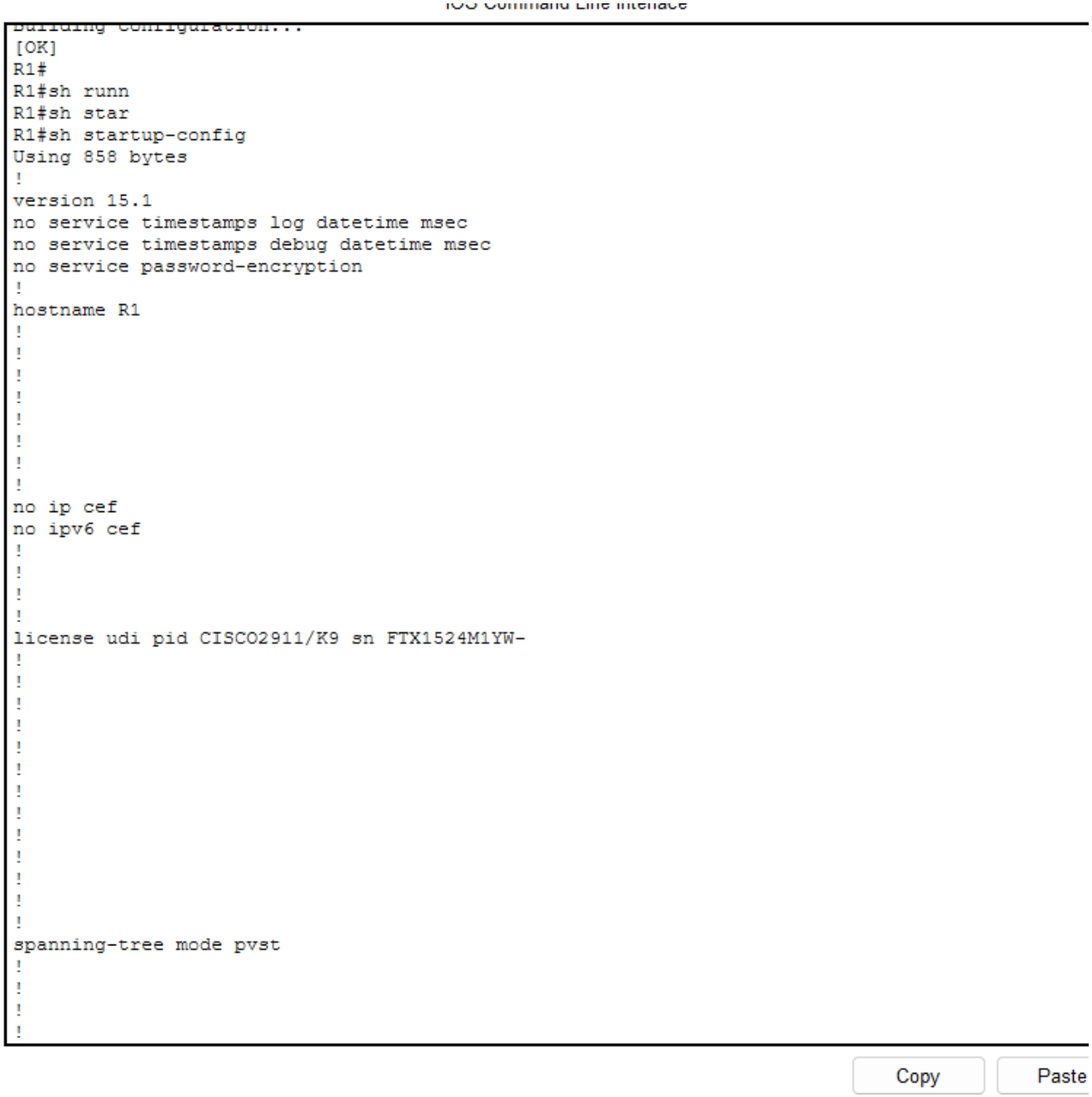


Figure 3. R1 saved configuration inspection (initial CLI output).

Figure 4. R1 final interface addresses and static/default route commands.

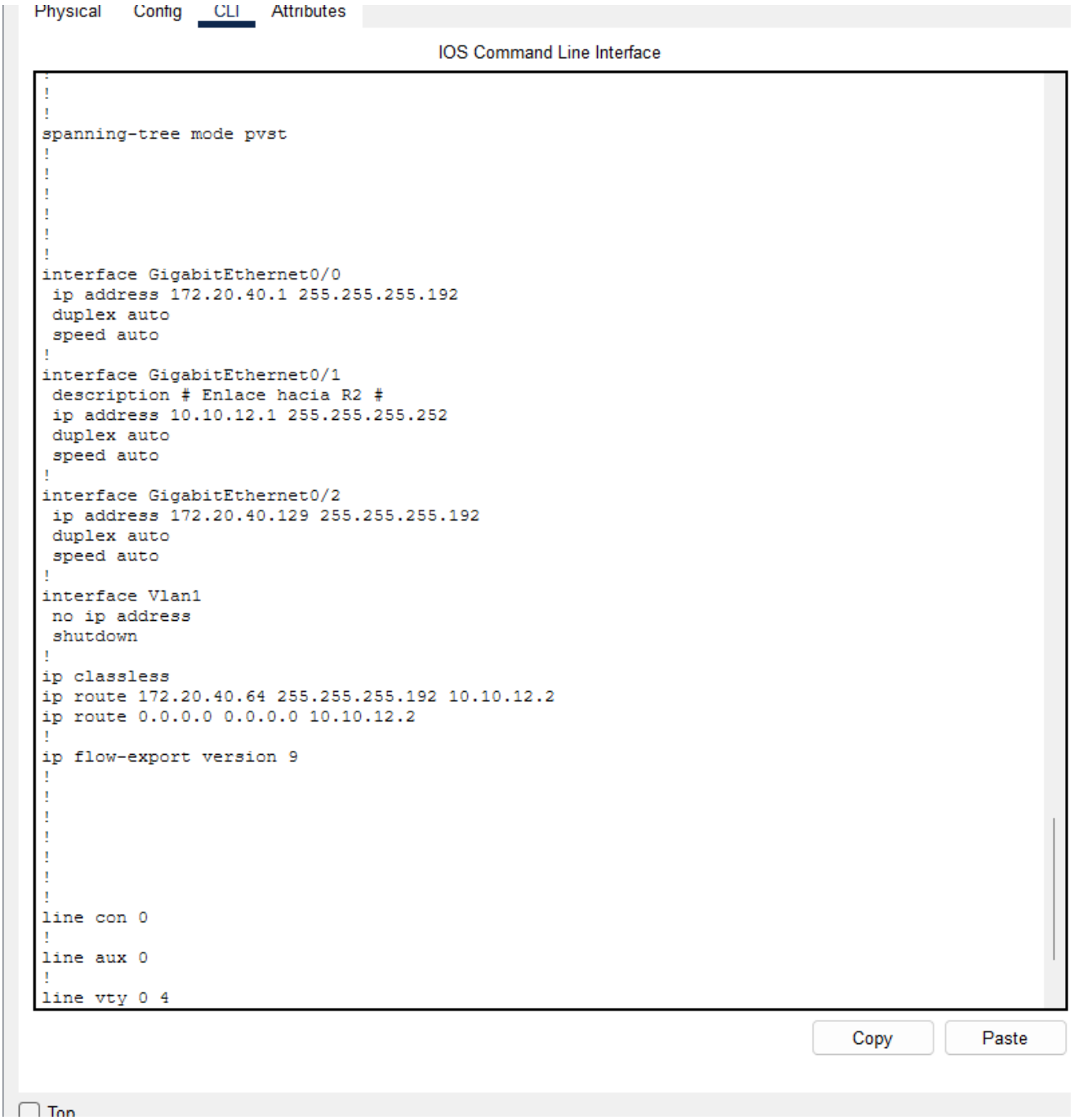


Figure 4. R1 final interface addresses and static/default route commands.

Figure 5. R1 routing table: connected, local, static, and default entries.

```
R1# sh ir oru
R1# sh ip rou
R1# sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 10.10.12.2 to network 0.0.0.0

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.10.12.0/30 is directly connected, GigabitEthernet0/1
L       10.10.12.1/32 is directly connected, GigabitEthernet0/1
     172.20.0.0/16 is variably subnetted, 5 subnets, 2 masks
C       172.20.40.0/26 is directly connected, GigabitEthernet0/0
L       172.20.40.1/32 is directly connected, GigabitEthernet0/0
S       172.20.40.64/26 [1/0] via 10.10.12.2
C       172.20.40.128/26 is directly connected, GigabitEthernet0/2
L       172.20.40.129/32 is directly connected, GigabitEthernet0/2
S*   0.0.0.0/0 [1/0] via 10.10.12.2

R1#
```

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Figure 5. R1 routing table: connected, local, static, and default entries.

Figure 6. R2 routing table and confirmation of saved startup configuration.

```
R2#sh ip rou
R2#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 10.10.12.1 to network 0.0.0.0

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.10.12.0/30 is directly connected, GigabitEthernet0/2
L       10.10.12.2/32 is directly connected, GigabitEthernet0/2
     172.20.0.0/16 is variably subnetted, 5 subnets, 2 masks
S       172.20.40.0/26 [1/0] via 10.10.12.1
C       172.20.40.64/26 is directly connected, GigabitEthernet0/0
L       172.20.40.65/32 is directly connected, GigabitEthernet0/0
C       172.20.40.192/26 is directly connected, GigabitEthernet0/1
L       172.20.40.193/32 is directly connected, GigabitEthernet0/1
S*   0.0.0.0/0 [1/0] via 10.10.12.1

R2#copy runn
R2#copy running-config star
R2#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
R2#
R2#
R2#sh star
R2#sh startup-config
Using 848 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
```

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Figure 6. R2 routing table and confirmation of saved startup configuration.

Figure 7. R2 final interface addressing and static/default route configuration.

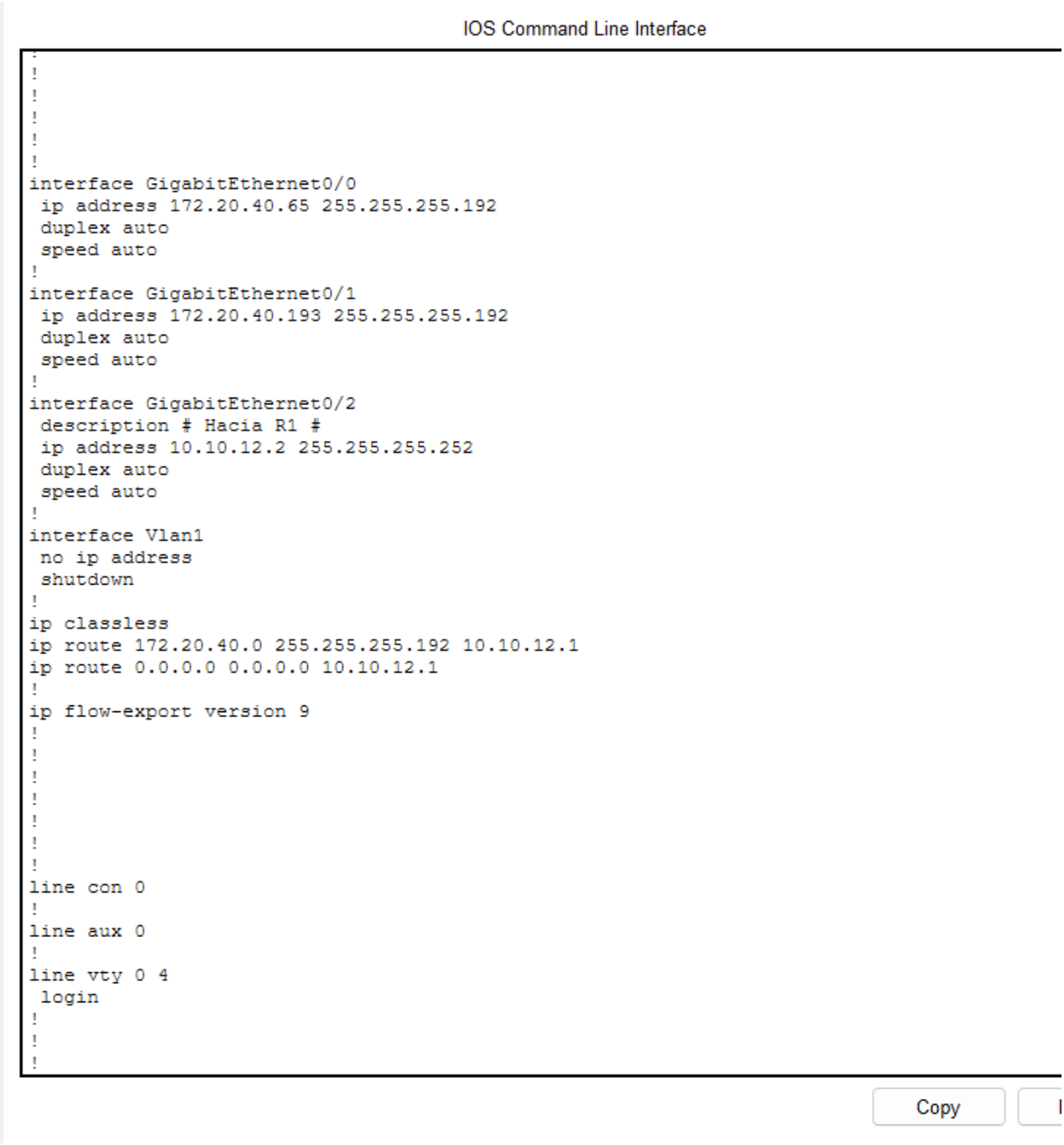


Figure 7. R2 final interface addressing and static/default route configuration.

Figure 8. PC7 successful reachability tests toward Engineering and Commercial.

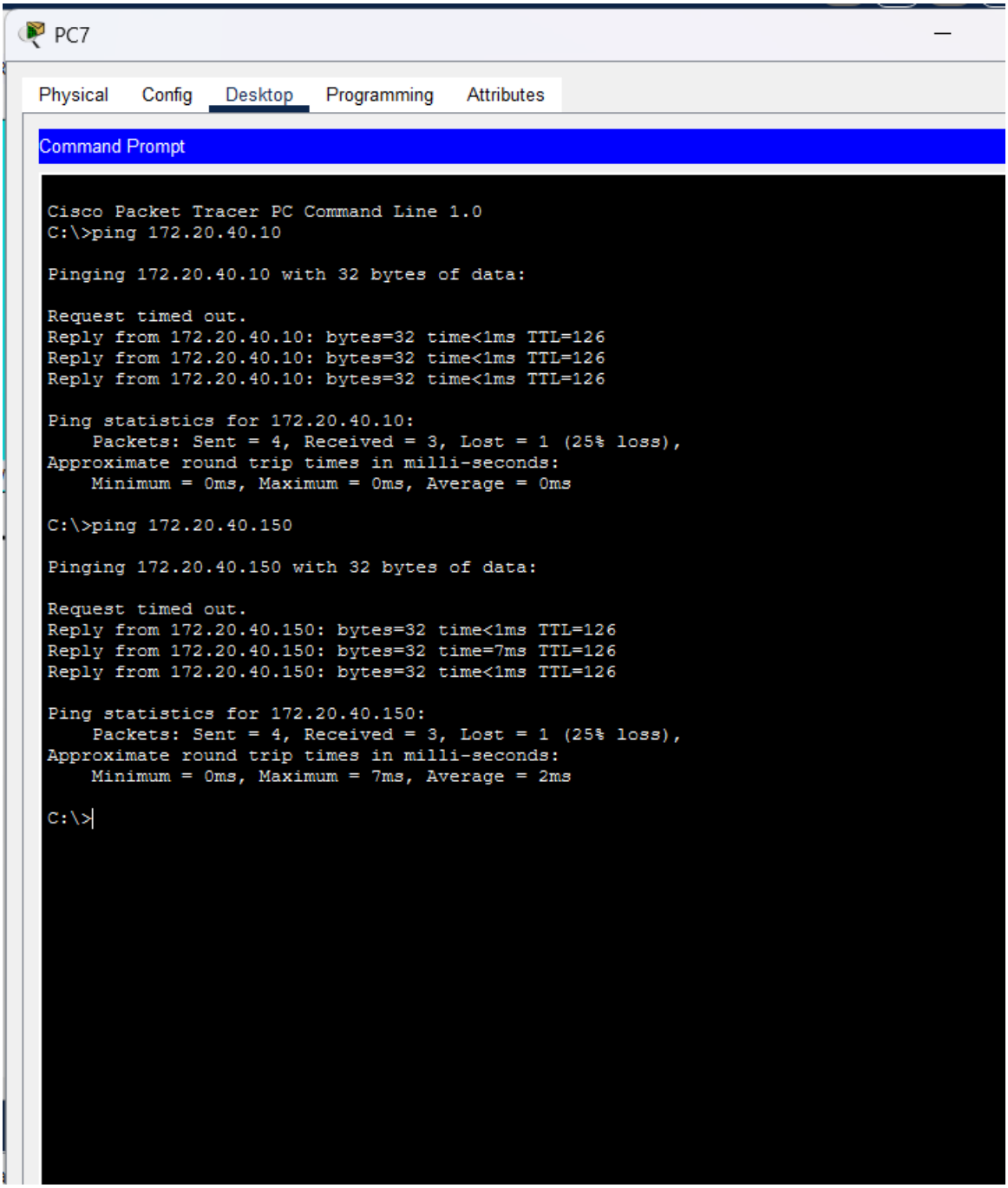


Figure 8. PC7 successful reachability tests toward Engineering and Commercial.